Property Rights, Firm Size and Investments in Innovation: Evidence from the America Invents Act

by

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Abstract

I analyze whether a change in patent systems differentially affects firm-level innovation investments at patent-valuing firms of different sizes. Using legally required, economically representative, U.S. Census Bureau microdata, I separate firms into groups based on a firm's response to a question asking it to rank the degree of patent importance to its business and firm-size. I then measure how firms' innovation inputs/outputs respond to the America Invents Act (AIA). Results show the AIA reduced innovation investments at smaller, patent-valuing firms while increasing innovation investments at larger, patent-valuing firms, highlighting differential firm-size effects of patent policy and policy's importance to investments.

Keyword: Investments, innovation, patents, firm size

JEL Classification: L25, O3, O51

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Does changing an economy's patent award system affect a firm's decisions to invest in and produce innovation? This is a fundamental question since property rights are critical to firms' investment decisions (Hart and Moore, 1990). Further, innovation is the primary driver of economic growth (Aghion, Van Reenen and Zingales, 2013). Notably, policies encouraging growth often target large firms, even though supporting small firms is more effective (Acemoglu et al., 2018). Hence, I examine whether changes to the U.S. patent system affect patent-valuing firms with different firm-sizes' investments in innovation inputs and outputs differently, focusing on the differences from the smallest to the largest, patent-valuing, firms.

My analysis examines the cumulative impact of the Leahy–Smith America Invents Act (AIA) which changed U.S. patent litigation in 2012[1] and the U.S.'s patent award process from first-to-invent to first-to-file in 2013. While patent reform was on the U.S.'s legislative agenda at least as early as 2005, it experienced numerous stalls and debates rendering when and how it would happen uncertain (Bartz, 2009). Pre-AIA, patent disputes were primarily litigated in district courts, and a patent was awarded to the earliest documented inventor. Post-AIA, new trial types (i.e., the Patent Trial and Appeal Board or PTAB) were introduced to reduce the time and costs associated with resolving patent disputes. The AIA also converted the U.S. to a first-to-file country: Post-AIA, a patent is awarded to the first inventor to file, even if said inventor was not the first to discover a technology. The U.S. was the last major economy to transition to a first-to-file patent system (Krishnamurthy, 2014). Critics of the U.S.'s transition argued it would place small firms at a disadvantage, leading to a decrease in small firms' innovative efforts (Abrams and Wagner, 2013). Advocates of the transition argued the first-to-file system would make the patenting process less complicated for all parties, including small firms (Gump and Smith, 2022). Determining which argument is correct is critical, especially for small firms. Not only are small firms instrumental to U.S. job creation (Haltiwanger, Jarmin and Miranda, 2013), but patents awarded to small firms "carve out" a technological space for future innovation (Galasso and

[1]The U.S. introduced the Patent Trial and Appeal Board (PTAB) in 2012 as part of the legislation that converted the patent system from first-to-invent to first-to-file.

Schankerman, 2015) and increase a startup's chance of securing venture capital funding, debt financing, or public financing via IPO. (Farre-Mensa, Hegde and Ljungqvist, 2020).

To address my research question, I use restricted, firm-level survey microdata, representative of the U.S. economy, from the U.S. Census Bureau's (Census) and the National Science Foundation's (NSF) Business R&D and Innovation Survey (BRDIS) from 2008 through 2016, with supporting datasets, in combination with the AIA as a natural experiment that changed the U.S.'s patent system. In my data, firms' survey responses are legally required, firms report the degree of utility patent importance to their businesses, and firms report information on investments in innovation inputs and outputs unavailable elsewhere. Thus, I analyze economically meaningful samples; mitigate survey selection bias; identify firms that are affected as a result of the AIA's enactment (i.e., firms that state utility patents are important to their businesses); and improve upon noisy innovation proxies in the extant literature by examining direct measures of firm-level (a) investments in innovation inputs and (b) resulting innovative outputs in a causal setting.

I examine the following investments in innovation inputs: (a.i) The rate of a firm's R&D expenditures *financed by the firm*; (a.ii) the percentage of R&D expenditures financed by a firm directed toward business areas or product lines that are new to the firm during the year; and (a.iii) a firm's rate of hiring R&D employees. I also examine the rate of hiring all employee types at a firm for comparison. The innovation outputs I analyze are: (b.i) The number of patents applied for at the United States Patent and Trademark Office (USPTO) during the year; (b.ii) the percent of USPTO patent applications a firm applied for, or plans to apply for, in foreign jurisdictions during the year; and (b.iii) the percent of sales from new or significantly improved goods and services that were new to a firm over the past three years.

To analyze the impact a change in patenting regimes has on patent-valuing firms with different firm-sizes' investments in innovation inputs and the resulting innovative outputs, I estimate separate difference-in-difference-in-differences (i.e., triple difference) models with a firm's investment in innovation inputs and resulting innovative outputs as regressands

(Olden and Møen, 2022). A firm is "treated" if it last reports prior to the AIA's enactment that utility patents are important to its business; whereas, a "control" firm last reports utility patents are not important to its business. I then compare how the innovation variables of treated and control firms respond to the AIA's enactment, and whether this difference between treated and control firms varies by a firm's size, measured by a firm's pre-AIA net sales and revenue. I find in all cases but one, the number of patent applications a firm files (i.e., output (b.i)), the smaller a treated firm is, the more it reduces its investments in innovation inputs and realizes fewer innovative outputs, post-AIA, relative to its control group. In contrast, very large, treated firms, post-AIA, increase investments in innovation inputs and realize increased outputs, relative to their control group. I infer the AIA caused smaller, patent-valuing firms to decrease investments in innovation and larger, patent-valuing firms to increase investments in innovation.

To validly draw causal inferences, I require a relatively weak identification assumption: I must assume in the absence of the AIA's provisions that the trend in the degree to which treated firms' investments in innovation inputs and resulting innovation outputs vary by firm-size would not have changed differently post-2012 than the trend in the degree to which control firms' investments in innovation inputs and resulting innovation outputs vary by firm-size. To confound identification, there would need to be an unrelated event that changes the degree to which a treated firm's investments in innovation inputs and resulting innovation outputs vary by firm-size differently than the trend in how a control firm's investments in innovation inputs and resulting innovation outputs vary by firm-size. While I cannot with certainty rule out the existence of such a confounding event, it is difficult to envision a sensible economic story that could rationalize one apart from the AIA itself since this was the only legislation affecting patenting during my analysis' time period from 2008 through 2016. An example of legislation that was concurrent with the AIA, and can be readily dismissed using my identification strategy, was the 2012 Jumpstart Our Business Startups Act (JOBS Act) (Congress.gov, 2012). The JOBS Act was aimed at relaxing financing constraints by making the securities registration process

less burdensome for small businesses. Since the JOBS Act would simultaneously affect both small, patent-valuing and small, non-patent-valuing firms' financing prospects, its effects would be removed via the second difference in my triple difference analysis. Moreover, I run placebo tests to determine whether some confounding event, not coinciding exactly with the AIA, could explain my results. I find for most innovation inputs/outputs the trends between treated and control firms begin to diverge in 2012 or later and not before. Thus, I can rule out the alternative hypothesis that my findings are attributable to a confounding shock occurring close to the AIA, but not in the same year.

Since a firm's survey response is required by law, I analyze the largest sample of firms possible for each BRDIS question to make full use of available data and avoid biasing results. The relative average treatment effect magnitudes I measure are substantial. In my largest sample, (a.i) the rate of a firm's R&D expenditures financed by the firm, my measurements imply a treated firm needs to have a pre-AIA firm-size equal to approximately \$64.6 billion to maintain their level of R&D expenditures equal to a control firm's, post-AIA. Hence, my measurements imply treated firms with pre-AIA firm-sizes greater than \$64.6 billion increase their R&D spending with respect to control firms' and treated firms with pre-AIA firm-sizes less than \$64.6 billion decrease their R&D expenditures relative to control firms', post-AIA.

Due to the substantial magnitude of pre-AIA sales needed for a treated firm to avoid a reduction in R&D expenditures, I separate firms into pre-AIA firm-size quintiles, to verify that smaller, treated firms are responsible for my results. Indeed, the difference-in-differences coefficients $Post$–$2012 \times Q1$ through $Q5$ are all positive for control firms, indicating investments in R&D among control firms increased in the post-AIA period, relative to the pre-AIA period. Whereas the estimated difference-in-differences coefficient for $Post$–$2012 \times Treat$ is negative and significant and the estimated triple difference coefficients $Post$–$2012 \times treat \times Q1$ through $Q5$ increase from significantly negative at the $Q1$ level to significantly positive at the $Q5$ level and support previous results. I also verify that a finer partition of my treatment group follows responses to the AIA that are consistent with

my previous results. Post-AIA, relative to control firms, firms that rank utility patents as "somewhat important" prior to the AIA decrease (a.i), the rate of R&D expenditures financed by the firm, less than firms that rank patents as "very important." Additionally, the triple difference coefficient for the "somewhat important" category is smaller in magnitude than the coefficient for the "very important" category.

Lastly, to bound my measurements of (a.i), I limit my sample to consist of firm-year observations where Census deemed a firm's yearly BRDIS form met completion criteria. In this smaller sample, estimates imply a firm needs to have a pre-AIA firm-size equal to approximately $21.8 million to maintain R&D expenditures the same as control firms', post-AIA. Since my largest sample decreases from $135,000$ to $91,000$ observations, I infer it is possible that the smaller a treated firm is the less likely it is to complete a survey and/or more likely to reduce its R&D expenditures to zero post-AIA.

Estimates of the pre-AIA firm-sizes necessary for treated firms to avoid decreases in my other measured innovation inputs are under $500 million each. I find for a treated firm to continue to maintain the same (a.ii), percentage of R&D expenditures financed by a firm directed toward business areas or product lines that are new to a firm during the year, as a control firm post-AIA requires a pre-AIA firm-size of approximately $405.9 million. My estimates imply for a treated firm to maintain the same (a.iii), rate of hiring R&D employees, as a control firm post-AIA requires a pre-AIA firm-size of approximately $89.9 million.

I observe similar relative reductions in treated firms' innovation outputs based on pre-AIA firm-size, post-AIA. Due to the staggered nature of innovation outputs, I estimate triple difference regressions with a full-set of pre/post year indicators. Coefficient estimates from my Poisson triple difference model for (b.i), the number of patents applied for at the USPTO during the year, show that a treated firm's incidence rate ratios for the difference-in-differences coefficients $I(Year) \times treat$ decline starting in 2014, after the first-to-file provision went into effect in 2013, relative to a control firm's. Whereas yearly, estimated triple difference coefficients, $I(Year) \times size \times treat$, are negative in 2013 and continue to

increase until they become positive in magnitude in 2016. This means, relative to their control group, that smaller, treated firms experienced higher relative incidence rate ratios for (b.i) prior to the first-to-file provision, and these incident rate ratios started to reverse after 2014. Interestingly, after 2015, all treated firms file fewer patent applications relative to their control groups. However, this effect is partially reduced for the largest firms in 2016.

My estimates for (b.ii), the percent of USPTO patent applications a firm applied for, or plans to apply for, in foreign jurisdictions during the year, show a similar trend to (b.i). Starting in 2012, relative to the base year of 2008, treated firms' (b.ii), declines by -9.14%, becomes slightly less negative in 2013 and continues to decline until it takes a value of -17.17% in 2016. The only way to mitigate these yearly declines is to have sufficient pre-AIA size, relative to the control group each year post-AIA. Lastly, for (b.iii), the percent of sales from new or significantly improved goods and services introduced during the previous three years that were new to a firm, a treated firm must have a pre-AIA size of approximately \$355.1 million to continue to follow the same trend as a control firm post-AIA.

My findings contribute to the extant literature in four main ways. First, I address the extant innovation literature's data limitations: My economically representative sample includes public and privately funded firms of varying firm-sizes and ages with as few as five employees. Second, I obtain novel, direct measures of firm-level investment in innovation inputs and outputs, even for small private firms. Third, I provide causal evidence that the process for awarding property rights via patents affects a firm's decisions to invest in R&D, hire and retain R&D employees, pursue new patents, and bring new products and services to market—and how these changes are a function of a firm's size. Fourth, this paper is the first to empirically establish that the AIA reduced smaller, patent-valuing firms' investments in innovation while increasing larger, patent-valuing firms' investments in innovation. My results support the conclusion that the design of a patent system plays a pivotal role in innovation at *all* firms, and that changes to said system may lead to

unintended policy effects.

I. Prior Literature and Contributions & Hypotheses

A. Prior Literature and Contributions

My analysis empirically tests the theoretical patent race literature with direct measures of firm-level investments in innovation inputs and outputs. My results support the predictions of the Miyagiwa (2015) model of patent races. Miyagiwa (2015) extended the Loury (1979) and Scotchmer and Green (1990) patent race models, predicting the AIA's transition from first-to-invent to first-to-file would decrease, or leave unchanged, a firm's R&D investments. Miyagiwa (2015) showed first-to-invent is generally more conducive to innovation. My results show a firm's size, relative to other firms, is a key factor in determining innovation investments. Specifically, firm-size differences, among firms with varying degrees of patent importance rankings for their businesses, can lead to either relative increases or decreases in innovation investments after a transition to first-to-file.

Further, I bypass previous data limitations of studies that used aggregated measures of investments, focused on selected firms' financials, and/or focused on patenting measures. My results are consistent with the aggregated measures of Lo and Sutthiphisal (2009), who analyzed Canada's 1989 change from first-to-invent to first-to-file at the industry-level.[2] Lo and Sutthiphisal (2009) concluded that while R&D expenditures did not change at the industry-level, patent ownership skewed towards large corporations. My results document a compositional shift in R&D spending, where small firms reduce the percent of R&D funds directed toward new business/product lines. Such compositional changes would remain undetected in aggregated NSF statistics documenting year-over-year increases in firms' R&D expenditures (Boroush, 2020). Further, my results are consistent with Abrams and Wagner (2013), who predicted that patenting would shift toward large firms/incumbents upon analyzing Canada's 1989 shift from first-to-invent to first-to-file.

Empirical studies examining the AIA are few and utilized datasets that did not capture

[2]The authors noted analysis at the firm-level was preferred, but experienced data unavailability.

a representative sample of firms. Lerner, Speen and Leamon (2015) verified Lo and Suthiphisal (2009), but did not find additional differences in tests on small and large firms in a stock market event study, or a decline in investments made by venture capital firms, following the AIA. The authors noted data availability was a concern and suggested studies comparing outcomes between small and large firms should be repeated at a later date. Chakravarty (n.d.) showed the AIA shifted firms' patenting behaviors among firms with higher exposure to litigation risk. Exposure to litigation risk resulted in a drop in follow-on innovation, measured by patents, among smaller firms/entities. Chakravarty (n.d.) is consistent with the reduction I observe in the rate of hiring R&D employees, post-AIA, at small firms. The hiring rate of R&D employees, those most likely to be responsible for innovation/product development, increased at large firms post-AIA relative to small firms. These employment dynamics are related to Akcigit and Goldschlag (2023) who showed U.S. inventors are increasingly concentrated in large incumbents. Further, Chakravarty's (n.d.) reduction in follow-on innovation is consistent with my result that smaller firms experience a reduction in the percent of sales from the introduction of new products/services.

Finally, both Huang et al. (2021) and Valentine, Zhang and Zheng (2023) examined the AIA among public firms via information disclosure. Huang et al. (2021) examined text-based measures of firms' 10-K R&D disclosures, concluding firms became more secretive when moving to first-to-file. Whereas, Valentine, Zhang and Zheng (2023) examined technological competition on innovation disclosures using public firms' patents post-AIA and found slower moving firms (i.e., "laggards") filed fewer patents and focused on fewer technologies, relative to rivals. Further, "laggard" firms strategically disclosed scientific publications to block rivals' patent applications. Similarly, I observe a reduction in patent applications and in the percent of applications slated for international filing concentrated among smaller firms.

Collectively, previous patent system studies were limited to aggregated variables, public firms, VC-backed firms, and patent/text-based measures, despite well-known differences in innovation related to firm-size and public/private status (Hall and Lerner (2010); Bernstein

(2015); Gao, Hsu and Li (2018); Driver, Kolasinski and Stanfield (2020); and Mezzanotti and Simcoe (2023)). My economically representative sample highlights that firms do not operate in isolation. Indeed, large firms interact with small firms to source innovation (Phillips and Zhdanov, 2013). My analysis also highlights the importance of small firms. In the U.S., small firms comprise approximately 99.9% of all firms, and were responsible for 65.1% of net new job creation since 2000 (U.S. Small Business Administration Office of Advocacy, 2020). Clearly, firms' patenting strategies change as a result of a change to a country's patent system. However, it remains unclear whether a change in patent regimes differentially impacts patent-valuing firms with different firm-sizes' investments in innovation inputs and outputs. My analysis fills this gap.

B. Hypotheses

Given the proximity of the PTAB's introduction in 2012, and the transition from first-to-invent to first-to-file in 2013, I consider their cumulative effects and call my hypotheses "AIA Hypotheses." To motivate my "AIA Hypotheses," I separately examine "PTAB Hypotheses" and "First-to-Invent to First-to-File Hypotheses," below, and arrive at the same two "AIA Hypotheses:" (H1) Small firms, unable to diversify risk over a greater number of assets/product offerings relative to large firms, reduce investments in innovation to be commensurate with risk and (H2) reductions in investment result in reductions in innovative outputs.

PTAB Hypotheses

One may argue the PTAB's introduction could benefit or harm small firms. On the one hand, the PTAB may benefit a small firm if a firm was sued in district court for patent infringement. Using the PTAB, an accused firm could file a petition against claims from the plaintiff's patent. If the accused firm won the AIA trial then said firm would be spared some litigation costs. Pre-AIA, the median costs, per side, for patent infringement litigation were $350,000 when less than $1 million was contested (Rajwani, 2010). On the other hand, the PTAB may disadvantage small firms. Legal expertise would still remain

expensive and small firms are less likely to have in-house legal counsel relative to large firms. Thus, while litigation costs may be reduced, it is ambiguous as to whether these cost reductions would allow small firms to navigate infringement lawsuits cheaper.

Critically, smaller firms likely own fewer patents. Thus, if invalidating a patent's claim(s) became easier, this would threaten a greater percent of a small firm's portfolio. Gill and Heller (2020) find valuable patent portfolios lead, on average, to 17% higher debt-to-asset-ratios. The authors showed results were driven by small, private firms. Additionally, Galasso and Schankerman (2015) found a patent's invalidation lead to a 50% decrease, on average, in patenting by the patent holder; further, this effect was entirely driven by small, innovative firms in technologies with many incumbents. By contrast, the effect of a large firm's patent being invalidated did not change the intensity of said firm's innovation. Post-invalidation, large firms pivoted R&D strategies. Less patent diversification increases the perceived risk of a small firm's investments in innovation relative to a large firm that can leverage diversification to decrease its innovative risk exposure (H1). Further, reductions in innovative inputs will lead to reductions in innovative outputs (H2).

First-to-Invent to First-to-File Hypotheses

The transition to first-to-file can also be argued to either benefit or harm small firms. Namely, a patent application date's cutoff plausibly exacerbates patent races. Schroth and Szalay (2010) showed "deep pockets" were associated with successful R&D performance, supporting the argument that large firms should benefit from intensifying patent races. Furthermore, the probable lack of in-house counsel at small firms lengthens the time it takes to file a patent application, and a firm may be hesitant to disclose information to a third party for fear of appropriation. The aforementioned will cause small firms to reduce investments commensurate with risk (H1) and decreases in investments will result in decreases in outputs for small firms (H2).

Both product development and distribution take time, and large incumbents have a head start. Large firms have access to more markets through superior product breadth/depth (Hoberg and Phillips, 2021). A small firm is unable to realize the same degree of product/sales diversification as a large firm. Hence, if a small firm realizes a loss in one of its product offerings, this likely represents a larger percent of its overall revenue. PTAB and the transition from first-to-invent to first-to-file lead to increased, non-diversifiable risk for small firms, which results in a reduction in investment commensurate with risk (H1) and results in fewer innovative outputs (H2).

II. Data

To test (H1) and (H2), I used Census' BRDIS and Longitudinal Business Database (LBD), in combination with Compustat to trace a firm's access to public financial markets, from 2008 through 2016, inclusive. BRDIS was an annual, firm-level survey jointly conducted by the Census and the NSF.[3] "[BRDIS] is the primary source of information on R&D expenditures and the R&D employees of for-profit, nonfarm businesses with five or more employees operating in the United States." (The National Science Foundation, 2020). A firm's survey response is required by law, and there are fines for providing false information. Surveyed firms are randomly sampled according to a probability proportional to size algorithm within a given strata based on R&D activity and NAICS codes from Census' Business Register. Firms with known R&D are sampled with a probability proportional to size based on R&D and firms with unknown R&D are sampled with a probability proportional to size based on annual payroll. Sampled firms with known R&D activity are sent the full-length BRDI-1 survey or "long form." Firms with unknown R&D performance are sent a "short form." A "short form" is a screener containing a subset of "long form" questions. Census sends the BRDIS even if firms do not have any R&D activity, so the sample allows for tests of treatment effects along both the intensive and extensive margins.

[3]Starting in 2008, BRDIS replaced the Survey of Industrial Research and Development.

Approximately 5,000 firms are sent the "long form" and approximately 40,000 firms are sent the "short form" each year (The National Science Foundation, 2020).

The LBD files used in this study are from 2008 through 2016. The LBD allows linking Census data, like annual BRDIS forms, to analyze firms longitudinally. The LBD comprises linked data for all employer establishments within the Census's Business Register and contains information on an establishment's payroll, employment, industrial activity, and age (Jarmin and Miranda, 2002).

Compustat was used to trace a firm's access to public debt and equity markets. A crosswalk between Census data and Compustat created an indicator variable for if a firm was part of Compustat in a given year. This indicator allows for restricting analyses to firms that access public financing in robustness tests to investigate firm-size effects.

A. Construction of the Dataset

Data consists of LBD linked BRDIS questionnaires from 2008 through 2016 to form a firm-year panel. Two indicator variables are merged into this panel to trace a firm's access to public financing: One for whether a firm is in the Compustat database each year and another for whether the firm is part of the Compustat database in 2011. Only firms that received the BRDIS at least once in the pre-AIA period (i.e., 2008 through 2011) and at least once in the post-AIA period (i.e., 2012 through 2016) are kept for analysis to ensure meaningful patent importance groups are utilized for comparisons. Further, prior to analysis, firms in the panel were categorized according to the mode of their yearly reported BRDIS NAICS codes (i.e., the mode of a firm's NAICS code from 2008 through 2016). In the event the mode of a firm's NAICS code could not be established, the firm was categorized according to the NAICS code it reported in its first observable BRDIS year. If a firm still did not have a distinct NAICS code for categorization it was removed from further analysis. The stratified nature of the BRDIS's sampling algorithm results in an unbalanced panel: Not every firm is consistently present in the panel from 2008 through 2016 or asked to complete the full length questionnaire. To make full use of available data, I investigate the largest sample of firms possible for each BRDIS question while

verifying/noting a question's availability across different form types in a given year.

B. Variables

Except for a firm's age from the LBD, and a firm's access to public financing from Compustat, BRDIS is the source of the variables to test Hypotheses (H1) and (H2). Below, I refer to questions from the 2008 and 2009 BRDI-1.[4] Care is taken to ensure question alignment across survey years and forms.

BRDIS Indicator Variables

Using the BRDIS-LBD panel, I construct three variables to estimate triple difference regressions. The first variable I construct is a year indicator for pre- or post-AIA. Pre-AIA is 2008 through 2011. Post-AIA is 2012 through 2016. The second variable I construct enables me to partition firms into pre-AIA patent importance categories. I use a firm's response to Question 6•10a., "During [YYYY], how important to your company were the following types of intellectual property protection? Utility patents (patents for invention)." This question has three check-box responses: "Very important," "Somewhat important," or "Not important." Since a firm's BRDIS response is legally required, I treat missing values as a firm answering utility patents are "Not important." I fix a firm's last response prior to 2012 in subsequent analyses.

The third variable I construct is a firm's pre-AIA size. To create pre-AIA size groups, I use a firm's last reported answer prior to 2012 to Question 1•6, "What was the amount of your company's worldwide net sales and operating revenues–the value of goods and services sold during the year, net of discounts, returns, and allowances–during [YYYY]?" If the question is unanswered, I check if this is filled in elsewhere on the survey since some additional questions require a respondent to copy previous answers to use in calculations to answer other questions. If a reported copy of Question 1•6's answer is missing, I use Question 1•7, "How much of the [YYYY] worldwide net sales and operating revenues reported

[4]The 2008 BRDI-1 is available at `https://www.nsf.gov/statistics/srvyindustry/about/brdis/surveys/srvybrdis_2008.pdf`. The 2009 BRDI-1 is available at `https://www.nsf.gov/statistics/srvyindustry/about/brdis/surveys/srvybrdis_2009.pdf`

in Question 1•6 was from domestic operations?" If this question is unanswered I check for carried over copies of Question 1•7 in other questions. I perform this categorization each year prior to 2012, and fix the last reported backwards looking value for subsequent analyses. Ultimately, if a firm failed to report any value in this backwards looking categorization, I set its size to zero sales. This method prevents firms with a potentially large size from being falsely categorized as small due to a single survey year, or error, and makes use of a BRDIS response being legally mandated.

BRDIS Innovation Investment Inputs

To test Hypothesis (H1), I use three BRDIS variables: (a.i) the rate of a firm's R&D expenditures financed by the firm during the year (i.e., asinh($R\&D$)); (a.ii) the percentage of R&D expenditures financed by a firm directed toward business areas or product lines that are new to the firm during the year (i.e., $R\&D$ *direction*); and (a.iii) a firm's R&D employment (i.e., asinh($R\&D$ *emp.*)). Concerning variable $R\&D$, I use BRDIS Question 2•1 "What was the total worldwide R&D expense for your company (including all domestic R&D expenses) in [YYYY]?" If the question is unanswered, I check if its value is filled in elsewhere on the survey since some additional questions require a respondent to copy previous answers to use in calculations to answer other questions. Otherwise, I set this value to zero. Notably, Question 2•1 is asked on both "short" and "long" forms each year.

Variable $R\&D$ enables me to improve upon past R&D measures since it excludes funding from outside sources. Using an R&D measure including external funding could hinder the interpretation of results due to two main issues. First, a grant or subcontract for R&D likely entails specific R&D objectives that may limit a firm's scope. Second, an outside source of R&D funding may be subject to agency problems: A firm's management may not face the same level of consequences when utilizing certain types of external capital (e.g., grants) as they would when risking a firm's own capital. Thus, I test a firm's willingness to put its investors' capital at risk pre- and post-AIA. Additionally, to make R&D expenditures comparable between firms, I take the inverse hyperbolic sine of $R\&D$, resulting in variable asinh($R\&D$). This is necessary since $R\&D$ is skewed and can be zero.

Concerning variable *R&D direction*, I examine a firm's answer to Question 3•4, "What percentage of the worldwide R&D expense in [YYYY] reported in Question 3•1 was directed toward business areas or product lines in which your company had no revenues before [YYYY]?"[5] This question is answered by firms who report R&D paid for by their firm greater than zero and is available on "long form" surveys. The variable *R&D direction* allows me to measure the proportion of R&D expenditures a firm is using to pursue new products/business lines as opposed to expenditures on marginal improvements to existing products. The variable *R&D direction* captures compositional shifts in a firm's R&D (i.e., variable *R&D*).

Variable *R&D direction* improves upon existing measures of innovative strategy that use granted patents for four main reasons. First, it is difficult to attribute R&D expenditures to particular items without detailed microdata. Second, a firm's innovative efforts may not be fully captured by patenting (Hall et al., 2014). Third, a firm's innovative activities may, or may not, result in a granted patent. Fourth, there is a well-known delay between when a firm applies for a patent and if/when said application is granted (Lerner and Seru, 2017). A decrease in *R&D direction* would support Hypothesis (H1) since it would show firms are moving away from business areas and product lines that are new to their company in response to the AIA. Whereas no change or an increase in *R&D direction* would suggest risks remained similar, or lessened, respectively, pre- versus post-AIA.

Next, I examine a firm's rate of hiring R&D employees, variable asinh(*R&D emp.*). Variable *R&D emp.* is available across form types. I use Question 5•2, "How many of your company's total worldwide employees reported in Question 5•1 were R&D employees?" If the question is unanswered, I check if this is filled in elsewhere on the survey since some additional questions require a respondent to copy previous answers. If unavailable, I use Question 5•6, "How many of your company's total domestic employees reported in Question 5•5 were R&D employees?" Due to skewness and the potential for this variable to be zero, I utilize the inverse hyperbolic sine of *R&D emp.*

[5]Question 3•1 is, "Copy the amount reported in Question 2•1–your company's worldwide R&D expense in [YYYY]."

The rationale for analyzing a firm's R&D employment to test Hypothesis (H1) is twofold. First, a firm's decision to retain or hire R&D employees represents a direct investment in innovation (Thompson, 2010). Second, a firm's R&D employees represent a stock of technical knowledge necessary to pursue future innovation (Hall and Lerner, 2010). Thus, decreases in R&D employment, post-AIA, would test if a firm reduces its investments in innovation. To interpret results, I further examine the rate of retention/hiring of all employees and this question is available across all survey forms and denote this variable as asinh($Emp.$).

BRDIS Innovation Outputs

To test Hypothesis (H2), I use three BRDIS variables: (b.i) Question 6•2, "How many patents did your company apply for in [YYYY] from the U.S. Patent and Trademark Office (USPTO)?"; (b.ii) Question 6•3, "What percentage of the patent applications reported in Question 6•2 has your company applied for or plans to apply for in foreign jurisdictions?"; and (b.iii) the sum of a firm's responses to Question 1-12a and 1-12b. Question 1-12a reads, "Using the definitions above, please give the percentage of your total sales in [YYYY] from: New or significantly improved goods and services introduced during [YYYY-2] to [YYYY] that were new to one of your markets?" and Question 1-12b. reads, "New or significantly improved goods and services introduced during [YYYY-2] to [YYYY] that were only new to your company?" Question 1-12a.'s and 1-12b.'s answers are constrained to sum to 100% when added to Question 1-12c.'s answer. Thus, the addition of Q-12a. and Q-12b. is valid to check for (b.iii) the percent of sales a firm realizes from new products/services during the previous three years (i.e., $t-2$ to t).

The number of patent applications a firm files at the USPTO, (b.i), is available on "long form" surveys and allows me to examine whether there is a change in a firm's likelihood to use patent applications to protect IP. The benefit of being able to directly observe the number of patent applications a firm files, instead of issued patents, is that applications are closer to when a firm invests in innovation and captures an investment in innovation even if a patent application does not result in a granted patent. If I observe a reduction

in a firm's innovation inputs, supporting Hypothesis (H1), a reduction in the number of patent applications would follow since a firm would be dedicating less resources to the inputs necessary to file a patent application with the USPTO.

To further aid in the interpretation of a firm's willingness to file patent applications, I examine (b.ii) the percent of patent applications filed at the USPTO slated for filing in foreign jurisdictions. This question is available on "long form" surveys. If a firm's management believes it has a valuable innovation, then it should seek to maximize profits by filing patent applications in as many jurisdictions as possible (e.g., Hall and Helmers (2019)). Thus, I expect to find there is no change in the percent of patent applications a firm plans to file for internationally among large, treated firms. If I found a difference favoring small, treated firms filing for more patent applications abroad, post-AIA, this would suggest small firms were producing innovations worthy of the costs of protection in foreign jurisdictions. This aids in interpreting results if one argues small, treated firms reduce patent applications to focus on breakthrough innovation.

Finally, starting in 2009, the year the BRDIS question becomes available, I analyze (b.iii) the percent of sales from a firm's new goods and services introduced during the past three years. This question is the sum of items a. and b. from Question 1-12 (referencing the 2009 BRDI-1, "long" form). This question is available across most survey forms and care is taken to ensure it is asked to both form groups. Hence, I do not analyze data from 2012 and 2013 in regressions since this question is not available across all survey forms in these years. However, I do report summary statistics for the largest sample possible.

My motivation for analyzing question (b.iii) is because it links Hypotheses (H1) and (H2). Granted patents from the United States Patent and Trademark Office (2018) specify "the right to exclude others from making, using, offering for sale, or 'selling' the invention in the United States or 'importing' the invention into the United States," and may or may not reflect a firm's goods/services available in the real economy: Depending on the industry, a specific product may require numerous patents to protect it from infringement. However, if I observe a decrease in innovation inputs, the reported percent of a firm's sales from

new goods and services allows me to investigate if these decreases translate to decreases in a firm's goods/service offerings in the real economy and highlights the importance of a patent system's structure such that a firm may earn a return on its investments in R&D.

Controls and BRDIS Metadata

I use the age of the firm's oldest establishment from the LBD to account for firm life cycle effects. The LBD starts in 1975, and the firm age variable is subject to truncation (Jarmin and Miranda, 2002). However, my analysis period is from 2008 to 2016, so truncation is expected to be minimal.[6] Further, to better describe the sampling structure of the BRDIS, I include indicator variables for a firm's inclusion in Compustat. The Compustat indicator allows me to observe whether a firm obtained financing from public markets. Thus, I can investigate innovation inputs and outputs among firms with public debt, or equity, in robustness tests. Additional variables include BRDIS metadata variables, such as an indicator specifying whether a firm-year observation met Census' internal requirements for form completion and an indicator for whether a firm-year observation received the "long form." The form completion indicator enables me to limit the sample of firms in robustness tests. The "long form" indicator enables me to limit samples to form-specific questions.

C. Summary Statistics

Table 1 presents the number of observations, mean, standard deviation, skewness, and kurtosis, for each variable. Table 1 is organized by "Panel A. Firm Characteristics," "Panel B. Innovative Investment Inputs," and "Panel C. Innovative Outputs."

[Insert Table 1 Here]

Under "Panel A," 74% of firm-year observations met Census survey completion. Thus, nonresponse does not present a major obstacle. Approximately 28% of observations were sent the "long form." About 12% of observations are in Compustat from 2008 through

[6] Initially, I report regressions with and without age controls and find minimal differences.

2016. Therefore, 88% of observations are private firms. The average firm-year net sales and revenue is $977 million. The average age of an observation is 24 years.

Upon grouping firms according to their last known net sales and revenue, prior to 2012, the average value is $996 million. Examining a firm's last known patent importance prior to 2012 establishes that 31% of firms answered patents are "Somewhat" or "Very" important. From this 31% total, 10% total reported patents are "Somewhat" and 21% total reported patents are "Very" important. Since firms are categorized by their last known patent importance response pre-AIA, this means 69% are categorized as having patents considered "Not important."

Under "Panel B," the average firm-year observation invested $19.7 million in R&D and had $2,578$ employees. Within a reduced sample, the average firm-year observation had 122 dedicated R&D employees. Furthermore, within the reduced sample of firms that received the "long form," an average of 20.2% of R&D expenditures were directed towards new areas and product lines (i.e., $R\&D\ direction$). Whereas under "Panel C," The average number of patents applied for is 14.5 and 22% of patent applications at the USPTO are, or planned to be, filed abroad. The percent of sales from new products/services introduced over the past three years is 17.7%.

III. Empirical Strategy

A. Hypothesis H1

To test innovation inputs hypothesized to change post-AIA, I separately estimate a triple difference model for each innovation input using the estimator from Correia (2017).

$$y_{igt} = \alpha_i + \lambda_{gt} + \beta_1 S_i P_t + \delta_1 P_t T_i + \delta_2 S_i P_t T_i + \mathbf{\Gamma}'\mathbf{X}_{igt} + \varepsilon_{igt} \tag{1}$$

In (1), i indexes a firm, g indexes the mode of firm i's six-digit NAICS code, and t indexes the year. Regressand, y_{igt} is either (a.i) $\text{asinh}(R\&D_{igt})$ to test the rate of a firm's R&D expenditures, (a.ii) $R\&D\ direction_{igt}$ to test the percent of a firm's R&D dedicated to the

introduction of new products/services, or (a.iii) asinh($R\&D\ emp_{igt}$) to test the rate of a firm's R&D employment. Additionally, y_{igt} is a firm's asinh($Emp.$) to test the rate of a firm's overall employment.

Variable P_t is a pre- and post-AIA indicator. The AIA was introduced in January 2011 and signed into law in September 2011. Its main provisions, the PTAB and the switch from first-to-invent to first-to-file, took effect in September 2012 and March 2013, respectively. I designate 2008 through 2011 to be pre-AIA, and 2012 through 2016 to be post-AIA. Variable P_t equals one if the year, t, is 2012 or later, and zero otherwise. Due to the AIA's introduction in 2011, and the progress legislation made toward passage over the course of the year, it is possible some effects were felt in 2011. I address this possibility in robustness tests.

Variable T_i is an indicator variable that identifies treated and control firms. To identify a treated and control firms, I use a firm's last reported answer to a BRDIS question, prior to 2012, asking a firm to rank the importance of utility patents to its business. A firm that last responds prior to the AIA's enactment that utility patents are "Somewhat important" or "Very important" is labeled treated and whose last response is missing or "Not important" is labeled as a control. Variable T_i, and is one for a treated firm i and zero otherwise.

Variable S_i is a firm-size measure. I take the inverse hyperbolic sine of a firm's last reported net sales and revenue pre-AIA and record this value as S_i for a firm i. Missing values are set to zero since the inverse hyperbolic sine is defined at zero, and a firm's BRDIS response is required by law. I use a firm's last known net sales and revenue, rather than payroll or employee count (the only other size measures available for most firms in my sample), because technological progress confounds interpretation.

Coefficients α_i are firm fixed effects and control for fixed, individual firm differences across time. Coefficients λ_{gt} are the mode of a firm's six-digit NAICS code by year fixed effects and control for industry-year shocks. Variable $\mathbf{X}_{igt}$ is a column vector of control variables, whose coefficients are estimated in row vector $\mathbf{\Gamma}'$. Variable $\mathbf{X}_{igt}$ includes the inverse hyperbolic sine of a firm's max. establishment age and its square. Variable ε_{igt} is

a random disturbance. Standard errors are clustered at the mode three-digit NAICS level from index g since (1) parametrically controls the time dimension with temporal mode six-digit NAICS by year fixed effects (Petersen, 2008).

Coefficient β_1 accounts for the difference in the control group's pre- and post-period firm-size trend: The fact that control firms may be investing in both innovation inputs and outputs as a function of pre-AIA size. Coefficient δ_1 captures the difference-in-differences estimate between treated and control firms in the pre-period relative to the post-period for firms with zero pre-2012 net sales and revenue. Coefficient δ_2 is the triple difference estimate and coefficient of interest in Hypothesis (H1) tests. Coefficient δ_2 accounts for the fact that treated firms with larger pre-AIA sizes are less likely to be financially constrained and should have greater financial flexibility to fund innovation and diversify innovative risks. The overall treatment effect is $\delta_1 + \delta_2 S_i$, scaling with a firm's pre-AIA size.

Further, in R&D robustness tests for (1), I disaggregate variable T_i into separate treatment effects for firms that answer patents are "Somewhat important" or "Very important" relative to "Not important." I also disaggregate firm-size effects into quintiles. Additionally, I separately examine all innovative input variables' year-by-year trends by including a full set of pre- and post-year indicators from 2009 to 2016, relative to a base year of 2008,[7] to verify the parallel trends assumption:

$$y_{igt} = \alpha_i + \lambda_{gt} + \left[\sum_{k=2009}^{2016} \beta_{1k} S_i I(k) + \delta_{1k} I(k) T_i + \delta_{2k} S_i I(k) T_i \right] + \mathbf{\Gamma}' \mathbf{X}_{igt} + \varepsilon_{igt} \qquad (2)$$

Covariates are the same as in (1), except $I(k)$. In (2), $I(k)$ is an indicator variable that equals one if the year is k and zero otherwise. If the parallel trends assumption is satisfied, the δ_{2k} coefficients should be statistically indistinguishable from zero before 2012.

[7] The base year is adjusted if a BRDIS question is not asked starting at the beginning of my sample in 2008.

B. Hypothesis H2

Due to the staggered nature of innovation outputs, I use a full set of pre- and post-year indicators as in (2). All other control variables remain the same. For continuous innovation output variables, I estimate the following equation using Correia's (2017) estimator:

$$y_{igt} = \alpha_i + \lambda_{gt} + \left[\sum_{k=2009}^{2016} \beta_{1k} S_i I(k) + \delta_{1k} I(k) T_i + \delta_{2k} S_i I(k) T_i \right] + \mathbf{\Gamma}' \mathbf{X}_{igt} + \varepsilon_{igt} \qquad (3)$$

Variable y_{igt} is either the (b.ii) percent of patent applications a firm plans to apply for in foreign jurisdictions or (b.iii) the percent of sales a firm realizes from new goods/services introduced during the past three years. If an output is a count variable, like (b.i) the number of patent applications a firm files during the year, I estimate (3) as a Poisson regression using the estimator from Correia, Guimarães and Zylkin (2020). In coefficient estimates, I should observe decreases in innovation inputs prior to decreases in innovation outputs. Hence, coefficients δ_{2k} in (3) should be indistinguishable from zero until 2013 or 2014 to allow time for both provisions of the AIA to be either fully anticipated or effective.

C. Identifying Assumptions

My strategy to test whether a change in patent systems causally affects patent-valuing firms with different firm-sizes' investments in innovation inputs and outputs differently is a triple difference extension of the continuous treatment difference-in-differences model analyzed in Callaway, Goodman-Bacon and Sant'Anna (2024) and used in Acemoglu and Finkelstein (2008). Hence, following an extension of the identification strategy used in Acemoglu and Finkelstein (2008), I must assume in my analysis that without the introduction of the AIA, patent-valuing (i.e., treated) firms with different firm-sizes would not have experienced differential changes in their investments in innovation inputs and outputs different from the differential changes in the investments in innovation inputs and outputs for non-patent-valuing (i.e., control) firms with different firm-sizes, post-2012. In other words, the differences between treated firms of different firm-sizes' investements in

innovation inputs and outputs should not trend differently from the differences in control firms with different firm-sizes' investments in innovation inputs and outputs in the AIA's absence. I empirically test my identifying assumptions by examining parallel trends as in Olden and Møen (2022). Additionally, I conduct sub-sample analyses related to firms' pre-AIA firm-sizes, treatment statuses, survey responses, and public/private statuses.

IV. Results

A. Innovation Investment Inputs

I report estimates of Equation (1) for R&D expenditures in Columns (1) and (2) of Table 2. Coefficients in Columns (1) and (2) are similar. Thus, as anticipated, the inclusion of firm age controls does not significantly alter estimates.

[Insert Table 2 Here]

The total treatment effect in Column (1) of Table 2 is the sum of the coefficients *Post–2012×treat* and *Post–2012×size×treat* multiplied by a firm's pre-AIA firm-size. Coefficient *Post–2012×treat* is -1.313 and significant at the 1% level. Coefficient *Post–2012×size×treat* is 0.0703 and significant at the 1% level. These estimates imply a transitional firm-size; meaning, post-AIA, relative to a firm's control group, treated firms below the transitional firm-size decrease their R&D expenditure rates and treated firms above the transitional firm-size increase their R&D expenditures rates.

The implied pre-AIA firm-size, from estimates in Column (1), to maintain the same R&D expenditure rate as a control firm post-AIA is approximately $64.6 billion. In percent terms, post-AIA, relative to a control firm with the same firm-size, using the 10.64 average firm-size value in Table 1 (approximately $20.9 million), and an approximation to the inverse hyperbolic sine, a treated firm decreases its R&D expenditure rate by approximately 43.2%. Similar results follow in Column (2) where the implied firm-size to maintain the same R&D expenditure rate as a control firm, post-AIA, is approximately $81.1 billion.

Both estimates in Columns (1) and (2) of Table 2 support hypothesis (H1). The smallest, treated firms reduce R&D expenditures and the largest, treated firms increase R&D expenditures relative to control firms, post-AIA. Since the firm-size estimates to avoid a substantial decline in R&D expenditure rates are large, I disaggregate firm-size and treatment variables in robustness tests in Table 3 to investigate further. I discuss my R&D expenditure rate robustness tests, with additional tests, below in "Innovation Input Robustness."

Estimated coefficients for a firm's *R&D direction* are in Column (3) of Table 2. A firm's *R&D direction* is only answered by "long form" firms with reported R&D expenditures greater than zero. Supporting results in Columns (1) and (2) of Table 2, the estimated coefficient for *Post–2012×treat* is -19.73% and the coefficient *Post–2012×size×treat* is 1.45%, scaling with the inverse hyperbolic sine of a firm's pre-AIA firm-size. Both estimates are statistically significant at the 5% level. The total treatment effect implies that, post-AIA, a treated firm must have a pre-AIA firm-size equal to approximately \$405.9 million to maintain the same percent of R&D expenditures budgeted towards new areas of business as a control firm of the same firm-size. This result further supports (H1).

Column (4) of Table 2 reports estimates for Equation (1) for a firm's *R&D employment* rate. The coefficient for *Post–2012×treat* equals -0.3025 and is significant at the 1% level. This negative coefficient further supports observed reductions in R&D expenditures and *R&D direction*. The estimate for *Post–2012×size×treat* equals 0.0250 and significant at the 1% level, indicating that treated firms with a larger pre-AIA firm-size increase R&D employment post-AIA, relative their control group; whereas smaller, treated firms decrease R&D employment, post-AIA, relative to their control group. Using an approximation, and the average value of firm-size reported in the Table 1, I estimate the average, treated firm experiences a reduction of approximately 3.6% in *R&D employment*, post-AIA, relative to its control group. The small percent reduction in the average *R&D employment* supports that firms are reluctant to release valuable human capital. Total treatment effect estimates imply that to zero out, or increase *R&D employment*, post-AIA, a firm's pre-AIA firm-size

needs to be approximately $89.9 million or greater, relative to a control firm's firm-size.

Lastly, for comparison, Column (5) of Table 2 reports estimates for Equation (1) for a firm's rate of hiring all employee types. Estimated coefficients for *Post–2012×treat* and *Post–2012×size×treat* are statistically insignificant. This indicates no statistically significant observable differences exist between treated and control firms for total employment. However, when taken in conjunction with the results in Column (4), these results indicate that the composition of treated firms' employees significantly changed post-AIA.

Innovation Investment Input Robustness

I organize robustness tests into two groups. In the first group, I test firms' R&D expenditure rates using four methods: (i) Sorting firms into pre-AIA firm-size quintiles; (ii) sorting firms into pre-AIA patent importance groups; (iii) limiting a regression to consist only of observations with satisfactory survey completion verified by Census; and (iv) limiting a regression to firms with entries in Compustat in 2011. I focus my robustness tests on R&D expenditures since it is available across BRDIS form types. I report estimates in Table 3. In the second group of robustness tests, I test parallel trends for each innovation input and report results in Table 4.

[Insert Table 3 Here]

Columns (1) and (2) of Table 3 report estimates, with and without firm age controls, respectively, for a firm's R&D expenditure rate after sorting firms into pre-AIA size quintiles. Due to the similarities in estimated coefficients' magnitudes and standard errors, I focus on results in Column (2). Estimates are reported relative to the omitted third firm-size quintile. Coefficient estimates for *Post–2012×Q1* through *Q5* are all positive, with the estimate for *Post–2012×Q5* significant at the 5% level. These quintile estimates show control firms with different firm-sizes all increased their R&D expenditures post-AIA, relative to the pre-AIA period.

However, the coefficient *Post–2012×treat* is -0.5791 and significant at the 1% level. Since a treated firm's total treatment effect scales with its pre-AIA firm-size, I note that as

a firm moves from the category *Post–2012×Q1×treat* (i.e., the smallest firms by pre-AIA sales) to *Q5* (i.e., the largest firms by pre-AIA sales), the sign of the triple interaction term changes from negative to positive after *Q3*. The coefficient estimated for *Post–2012×Q1×treat* is −0.5078 and significant at the 1% level and the coefficients estimated for *Post–2012×Q4×treat* and *Q5* are 0.1573 and 0.2130, respectively, and significant at the 10% and 5% levels, respectively. This indicates that, post-AIA, relative to control firms, it is only the largest, treated firms in my estimated models with continuous firm-size that are able to avoid decreases in R&D expenditures. Thus, my firm-size quintile results confirm it is the smallest, treated firms, in my continuous firm-size model, that are responsible for my results.

Columns (3) and (4) of Table 3 examine a firm's R&D expenditure rate after separating the pre-AIA patent importance variable into "Somewhat" and "Very" important categories, relative to a "Not important" base. Column (4) includes firm age controls, whereas Column (3) does not. Due to the similarities in estimated coefficients, I discuss estimates in Column (4). In Column (4), the coefficients estimated for *Post–2012×very important* and *Post–2012×somewhat important* are both negative and statistically significant at the 5% and 1% level, respectively. I note the ratio of the magnitude of *Post–2012×very important* relative to *Post–2012×size×very important* is approximately equal to 17.7 whereas the ratio of the magnitude of *Post–2012×somewhat important* to *Post–2012×size×somewhat important* is approximately equal to 40.

These differences in ratios between the "Somewhat" and "Very" important categories have meaningful consequences for the implied pre-AIA firm-size a firm must have to avoid reducing R&D expenditures relative to a control firm, post-AIA. Specifically, the total treatment effect for the *very important* coefficients implies a "Very important" treated firm needs to have a pre-AIA firm-size of approximately $24.7 billion to maintain the same rate of R&D investment as a control firm, post-AIA. Whereas the coefficients for *somewhat important* imply a pre-AIA firm-size magnitude such that no firm rating patents as "Somewhat important" would be able to avoid reducing R&D expenditures relative to

a control firm, post-AIA. Therefore, these results show the change in patent systems has a "clearing" effect. Post-AIA, relative to their control group, only firms that rank utility patents as "Very important" to their businesses, with sufficient pre-AIA firm-size, are able to avoid the largest R&D divestments or increase R&D investments. Whereas, firms that rank utility patents as "Somewhat important" to their businesses, dramatically decrease R&D expenditures relative to control firms, post-AIA.

In Column (5) of Table 3, I limit the R&D expenditures sample to firms Census indicated adequately completed the BRDIS in a given year. The sample decreases from $130,000$ to $91,000$ observations. Estimates for *Post–2012×treat* and *Post–2012×size×treat* decrease in magnitude but both remain statistically significant at the 5% level. These estimates imply that post-AIA a treated firm needs to have a pre-AIA firm-size equal to $21.8 million to maintain the same R&D expenditure rate as a control firm. Due to the significant decrease in sample size, I infer that it is possible that the smaller a firm is, the less likely it is to remain in sample. Therefore, I interpret $21.8 million as the lower bound on the treatment effect size for a treated firm to maintain its R&D expenditure rate the same as a control firm, post-AIA.

In untabulated results, I limit the sample to consist only of firms with observations in Compustat in 2011 to trace a firm's access to public financing. I estimate Equation (1) with firm age controls for a firm's rate of R&D expenditures as in Table 2, Column (2). I find the coefficient *Post–2012×treat* is still negative but only significant at the 10% level. Importantly, I find all size-related coefficients are statistically insignificant. The same result follows when testing *R&D direction* in the public financing sample, as in Column (3) of Table 2: Coefficient *Post–2012×treat* is negative and significant at the 10% level whereas *Post–2012×size×treat* is statistically indistinguishable from zero. Additionally, when repeating the analysis of the rate of R&D employee hiring and the rate of all employee hiring, as in Columns (4) and (5) of Table 2, utilizing the public sample, coefficients *Post–2012×treat* and *Post–2012×size×treat* are statistically indistinguishable from zero in both estimates. These results indicate that small, private firms are responsible for my results.

Further utilizing the public financing sample, I re-estimate quintile results in Table 3, Column (2). Except for a positive and statistically significant coefficient for *Post–2012×Q1*, I observe no other statistically significant effects related to treatment or size. This supports earlier results that small, private firms are driving estimates. I also re-estimate the treatment category results in Column (4) of Table 3 using the public financing sample and find only firms that ranked patents as "Very important" experience any effects. The coefficient for *Post–2012×very important* is negative and statistically significant at the 5% level. While the coefficient for *Post–2012×size×very important* is positive and statistically significant at the 10% level. I also further limited the public financing sample to consist of firms determined to have adequately completed the BRDIS in a given year, as in Table 3 Column (5), and estimates of treatment and size coefficients are statistically indistinguishable from zero.

Next, I test the parallel trends assumption for each innovation input by estimating Equation (2). I report results in Table 4.

[Insert Table 4 Here]

Columns (1) through (3) of Table 4 examine a firm's R&D expenditure rate. Due to the estimated coefficients' similarities in Columns (1) and (2), I focus on Column (2) and compare its estimates with those in Column (3). Column (3) limits my sample to consist of firms with satisfactory survey completion determined by Census. The triple difference coefficient estimates in Column (3), $I(YYYY) \times size \times treat$, satisfy the parallel trends assumption and contain a positive and statistically significant triple interaction effect in 2015. Whereas the triple difference coefficients in Column (2), do not satisfy the parallel trends assumption. Despite triple difference estimates' violation of the parallel trends assumption in Column (2), these estimates support my earlier inference that it is the smallest, treated firms that are responsible for my results. Further, it is important to analyze potential reasons for the parallel trends' discrepancy between Columns (2) and (3). Thus, with respect to Columns (2) and (3), I first discuss the signs and magnitudes of the yearly difference-in-differences coefficients, $I(YYYY) \times treat$, and then discuss the

signs and magnitudes of the yearly triple difference coefficients, $I(YYYY) \times size \times treat$. I then examine the parallel trends assumption for my remaining innovation inputs: *R&D direction* and a firm's hiring rate of R&D employees compared with a firm's hiring rate for all employee types.

Upon examining estimates in Column (2), there is a large decrease in magnitudes for the coefficients $I(YYYY) \times treat$ coinciding with the same year of the AIA's first provision in 2012 and continuing through 2016. In Column (2), starting in 2009 for $I(YYYY) \times treat$, treated firms are 0.3182 higher in average R&D expenditure rates than in 2008 and the yearly estimated coefficient value continues to rise until it reaches a maximum of 1.4305 in 2011. Interestingly, in 2012, the coefficient $I(YYYY) \times treat$ changes sign to -0.1025 and remains negative throughout the remainder of my sample, decreasing to a low of -1.0564 in 2015 and remaining negative at -0.8110 in 2016. A similar trend is observed in Column (3), where yearly difference-in-differences coefficients, $I(YYYY) \times treat$, are 0.6186 starting in 2009 and increase to a value of 0.9969 in 2011. Then, starting in 2012, the value decreases to 0.4175 and tapers down to 0.2335 in 2016. Thus, both Column (2) and (3) contain a large drop in a treated firm's rate of R&D expenditures starting precisely in the year of the AIA's first provision. Due to the fact that these estimates are for firms with zero pre-AIA firm-size, it supports the inference that it is the smallest firms in Column (2) that are driving my coefficient estimates to be negative for $I(YYYY) \times treat$ since said firms are either cutting R&D expenditures to zero, not filling out the survey completely, or are engaged in some combination of these two extremes. However, since the survey is required by U.S. law and designed to be a representative sample fo the U.S. economy, I reasonably infer that most treated, small firms are cutting R&D expenditures to zero with a smaller fraction not meeting Census' definition of survey completion.

Opposite the decreasing pattern in yearly estimates for the difference-in-differences coefficients, $I(YYYY) \times treat$, the triple difference coefficients, $I(YYYY) \times size \times treat$, in Column (2), exhibit a large, positive change in magnitude starting in 2012. Specifically, the coefficient, $I(2012) \times treat$, rises from 0.0568 in 2011 and takes the value 0.1406 and

remains elevated through 2016. Column (3) exhibits a similar pattern. Namely, the triple interaction terms in Column (3) are estimated near zero prior to 2012. Starting in 2012 and later, they are consistently positively estimated. This indicates that treated firms with larger pre-AIA size are increasing R&D investments post-AIA.

The same coefficient pattern for the yearly difference-in-difference and triple difference coefficients emerge for my final two innovation inputs *R&D direction* and a firm's rate of hiring R&D employees. The parallel trends assumption holds for *R&D direction* in Column (4). Relative to the base year of 2008, treated firms decrease the percentage points of R&D budgeted to new products/services by -19.81% starting in 2012 and sustain this large percent decrease through 2016. The triple interaction terms show a sustained positive and significant increase of approximately 1.35% points in 2012 that scales by multiplication with the inverse hyperbolic sine of a firm's pre-AIA size and increases to a value of 2.05% points in 2016.

Further supporting results in Columns (3) and (4), the rate of a firm's R&D employee hiring in Column (5) trends negative for treated firms post 2012 through 2016. Whereas the triple interaction term becomes positive and significant post 2012 through 2016. Thus, larger, treated firms are hiring more R&D employees than their smaller, treated counterparts post-AIA, when compared to their respective firm-size counterfactual. Lastly, Column (6) provides an interesting comparison with Column (5). The parallel trends assumption does not hold, but it shows small, treated firms hire at faster rates than their large, treated counterparts, supporting the estimate in Column (5) of Table 2 for coefficient *Post–2012×size×treat*.

B. Innovation Outputs

Table 5 reports estimates for Equation (3) for a firm's innovation outputs. Column (1) contains estimates for the number of a firm's yearly patent applications with the USPTO. Column (2) contains estimates for the percent of patent applications that are considered for international filing. Column (3) contains estimates for the percent of sales over the past three years from the introduction of new products and/or services. Lastly, Column

(4) contains period collapsed pre- and post-2012 estimates for a firm's percent of sales from new products and/or services over the past three years.

[Insert Table 5 Here]

Table 5 shows that small firms filed fewer patent applications at the USPTO (i.e., Column (1)) and of those filed, fewer of these applications were considered for international filing except those at larger firms (i.e., Column (2)). The parallel trends assumptions hold in both Columns (1) and (2). Lastly, results in Columns (3) and (4) indicate that the AIA has a very significant effect on firms' products that appear in the real economy: Firms realize very different sales compositions pre-AIA vs. post-AIA. Specifically, firms are introducing fewer novel products unless they are larger firms. The average treated firm's composition of sales from new products/services declines by 15.17% and remains negative through 2016. Further, this stark decline is only avoided by firms with large enough size, as indicated by the positive and statistically significant coefficients on $I(YYYY) \times size \times treat$.

V. Conclusion

The effect of a change in patent systems has markedly different consequences for firm-level investments in innovation depending on a firm's size. Utilizing the AIA as a shock to firm-level innovation, I showed firm's that rated utility patents as important to their businesses (i.e., treated firms) shifted their innovation strategies on the basis of firm-size relative to control firms, post-AIA. The smallest, treated firms divested from R&D expenditures, decreased R&D employment, and shifted their R&D investments from novel to existing goods/services, post-AIA. Whereas, the largest, treated firms increased their R&D investments, R&D employment, and the percent of R&D spending dedicated towards new products/services, post-AIA. As a result of a decrease in innovation inputs, the smallest, treated firms filed fewer patent applications and realized less sales from new goods/services, while the largest, treated firms realized the opposite outcomes in all but the case of the number of patent applications filed each year.

A proposed mechanism driving my results is that small, treated firms cannot diversify innovation risks as effectively as large, treated firms: Losing one patent is not as critical to a large firm as it is to a small firm (Galasso and Schankerman, 2015) and small firms do not have the "deep pockets" necessary to compete in patent races (Schroth and Szalay, 2010). Both of the aforementioned risks, respectively, coincide well with the AIA's introduction of the PTAB, a low-cost channel to challenge patents, and the AIA's transition from first-to-invent to first-to-file. My results show: The AIA's reforms tilted innovation in favor of large, treated firms, resulting in measurable effects on firm-level investments and the real economy; and the design of a patent system plays a critical role in fostering innovation at all firms and has the power to change the composition of firms supplying innovation.

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Table 1—Summary Statistics (Firm-Level)

	Observations	Mean	Standard deviation	Skewness	Kurtosis
Panel A. Firm Characteristics					
BRDIS response (indicator)	135,000	0.7380	0.4397	−1.083	2.172
BRDI-1 "long form" (indicator)	135,000	0.2824	0.4502	0.9669	1.935
Compustat (indicator)	135,000	0.1176	0.3221	2.374	6.637
Net sales & rev. (000s USD)	135,000	976,700	8,176,000	31.48	1,699
Net sales & rev., 2011 or earlier (000s USD)	135,000	996,400	8,347,000	28.58	1,246
Size (asinh(000s USD))	135,000	10.64	3.560	−1.317	5.718
Treat (indicator)	135,000	0.3080	0.4617	0.8316	1.692
"Somewhat important" (indicator)	135,000	0.09888	0.2985	2.688	8.223
"Very important" (indicator)	135,000	0.2092	0.4067	1.430	3.046
Max. estab. age (years)	131,000	24.01	11.85	−0.3125	1.687
asinh(Max. estab. age) (asinh(years))	131,000	3.675	0.7358	−1.418	5.041
asinh^2(Max. estab. age) (asinh^2(years))	131,000	14.05	4.704	−0.8487	2.685
Panel B. Innovative Investment Inputs					
R&D (000s USD)	135,000	19,730	242,600	28.28	1,007
asinh(R&D) (asinh(000s USD))	135,000	3.812	4.352	0.5087	1.739
R&D direction (percent)	23,000	20.18	31.70	1.577	4.117
R&D emp. (count)	110,000	122.3	1,080	23.05	683.6
asinh(R&D emp.) (asinh(count))	110,000	2.186	2.258	0.7540	2.903
Employment (count)	135,000	2,578	23,000	68.20	7,335
asinh(Employment) (asinh(count))	135,000	5.906	1.983	0.6935	3.409
Panel C. Innovative Outputs					
Patents applied at USPTO (count)	75,000	14.45	396.7	217.0	54,050
Patents applied at USPTO planned for international application (percent)	51,000	21.88	37.88	1.341	3.011
Percent sales new to market/firm $t-2$ to t (percent)	56,500	17.68	27.68	1.855	5.431

Note: Estimates adhere to Census requirements. Firm-Year observations are from 2008–2016, unless otherwise specified. A variable's unit of measurement, or reference unit for dimensionless variables, is shown in parentheses with its denominator (i.e., year) omitted for tractability. USD values are not CPI adjusted. Variables are from BRDIS, LBD, and Compustat. The sample requires every firm to receive the BRDIS questionnaire at least once in 2011, or earlier, and at least once in 2012 or later. Variable *Size* is asinh(Net sales & rev., 2011 or earlier). If a firm fails to report net sales and revenue from 2011, or earlier, its size is set to zero. Variable *Treat* is an indicator that equals one if a firm's last reported utility patent importance is "Somewhat" or "Very" important, as of 2011 or earlier, and zero otherwise. If a firm fails to report patent importance in 2011 or earlier, it is categorized as answering patents are "Not important" and placed in the control group. Variable *"Somewhat important"* is one if a firm's last reported utility patent importance is "Somewhat important," as of 2011 or earlier, and zero otherwise. Variable *"Very important"* is one if a firm's last reported utility patent importance is "Very important," as of 2011 or earlier, and zero otherwise. Variable *R&D direction* is the percent of R&D directed towards new areas/products.

Table 2—Innovation Investment Inputs (Firm-Level)

	asinh(R&D)	asinh(R&D)	R&D direction	asinh(R&D emp.)	asinh(Emp.)
	(1)	(2)	(3)	(4)	(5)
Post–2012 × *size*	0.0219	0.0220	−0.38	−0.0094	0.0004
	(0.0119)	(0.0110)	(0.69)	(0.0052)	(0.0021)
Post–2012 × *treat*	−1.313	−1.259	−19.73	−0.3025	0.0043
	(0.139)	(0.123)	(7.74)	(0.0876)	(0.0256)
Post–2012 × *size* × *treat*	0.0703	0.0666	1.45	0.0250	−0.0037
	(0.0123)	(0.0106)	(0.64)	(0.0075)	(0.0023)
asinh(*Max. estab. age*)		−0.2104	−8.67	0.0630	0.1578
		(0.0941)	(6.59)	(0.0520)	(0.0621)
asinh^2(*Max. estab. age*)		0.0296	1.09	0.0002	−0.0029
		(0.0197)	(1.28)	(0.0106)	(0.0128)
# clusters	80	80	60	80	80
Adj. R^2	0.622	0.626	0.411	0.778	0.954
Obs.	135,000	130,000	17,500	102,000	130,000

Note: Estimates adhere to Census requirements. Triple difference regression coefficients and standard errors are reported from regressions using BRDIS panel data from 2008–2016. All regressions include firm fixed effects and six-digit mode NAICS by year fixed effects. Standard errors are clustered at the three-digit NAICS level and calculated from disclosed t-statistics. Regressands in Columns (1)–(5) are defined in "Variables." Column (3) is limited to BRDI-1 forms. The variable *Size* is the inverse hyperbolic sine of a firm's last known net sales and revenue from 2011, or earlier. Variable *Size* is taken to be zero if no value is reported from 2011 or earlier. The variable *Treat* is an indicator that equals one if a firm's last known response in 2011, or earlier, was that utility patents are "Somewhat important" or "Very important" and zero otherwise (i.e., patents are "Not important"). The variable *Post–2012* is an indicator that equals one if the year is 2012 or later, and zero otherwise. Firm-year observations with *Net Sales and Rev.* < 0, *R&D* < 0, or *Emp.* ≤ 4 are dropped from estimates.

Table 3—R&D Firm-Size, Treatment, & Sample Robustness (Firm-Level)

	Size		Treatment		Sample
	asinh(R&D)	asinh(R&D)	asinh(R&D)	asinh(R&D)	asinh(R&D)
	(1)	(2)	(3)	(4)	(5)
Post–2012 × $Q1$	0.0901	0.0937			
	(0.1308)	(0.1228)			
Post–2012 × $Q2$	0.0662	0.0649			
	(0.0523)	(0.0546)			
Post–2012 × *size*			0.0218	0.0217	0.0187
			(0.0121)	(0.0111)	(0.0091)
Post–2012 × $Q4$	0.0679	0.0635			
	(0.0520)	(0.0542)			
Post–2012 × $Q5$	0.1702	0.1709			
	(0.0679)	(0.0703)			
Post–2012 × *treat*	−0.6211	−0.5791			−0.3397
	(0.0781)	(0.0764)			(0.1673)
Post–2012 × *somewhat important*			−0.7686	−0.6833	
			(0.2955)	(0.2832)	
Post–2012 × *very important*			−1.461	−1.412	
			(0.141)	(0.129)	
Post–2012 × $Q1$ × *treat*	−0.4720	−0.5078			
	(0.1679)	(0.1642)			
Post–2012 × $Q2$ × *treat*	−0.0551	−0.0858			
	(0.1601)	(0.1844)			
Post–2012 × *size* × *treat*					0.0318
					(0.0140)
Post–2012 × *size* × *somewhat important*			0.0259	0.0171	
			(0.0256)	(0.0240)	
Post–2012 × *size* × *very important*			0.0817	0.0797	
			(0.0130)	(0.0112)	
Post–2012 × $Q4$ × *treat*	0.1714	0.1573			

Continued on next page

Table 3 – *Continued from previous page*

	Size		Treatment		Sample
	asinh(R&D)	asinh(R&D)	asinh(R&D)	asinh(R&D)	asinh(R&D)
	(1)	(2)	(3)	(4)	(5)
	(0.0860)	(0.0927)			
$Post$–2012 $\times$ $Q5$ $\times$ $treat$	0.2631	0.2130			
	(0.1123)	(0.1066)			
asinh($Max.\ estab.\ age$)		−0.2606		−0.2081	0.0905
		(0.0913)		(0.0943)	(0.0947)
asinh^2($Max.\ estab.\ age$)		0.0367		0.0293	−0.0132
		(0.0194)		(0.0198)	(0.0249)
# clusters	80	80	80	80	80
Adj. R^2	0.622	0.626	0.622	0.626	0.803
Obs.	135,000	130,000	135,000	130,000	91,000

Note: Estimates adhere to Census requirements. Triple difference regression coefficients and clustered standard errors are reported from regressions using BRDIS panel data from 2008–2016. All regressions include firm fixed effects and six-digit mode NAICS by year fixed effects. Standard errors are clustered at the three-digit NAICS level and calculated from disclosed t-statistics. Definitions of regressands are in "Variables." Column (5) limits the sample by an internal Census form completion indicator. Variables *Q1–Q5* are size quintiles based on the inverse hyperbolic sine of a firm's last reported net sales and revenue from 2011 or earlier (denoted as *Size*). Variable *Size* is taken to be zero if no value is reported from 2011 or earlier. Variable *Q1* is $I(Size \leq p20)$, *Q2* is $I(p20 < Size \leq p40)$, *Q3* is the omitted reference quintile, *Q4* is $I(p60 < Size \leq p80)$, and *Q5* is $I(p80 < Size \leq p100)$, where $I()$ is an indicator function that equals one if the argument is true and zero otherwise. The variable *Treat* is an indicator that equals one if a firm's last known response in 2011 or earlier, is that utility patents are "Somewhat" or "Very" important and zero otherwise (i.e., patents are "Not important"). The variable *Post–2012* is an indicator variable that equals one if the year is 2012 or later, and zero otherwise. Firm-year obs. with $Net\ Sales\ and\ Rev. < 0$, $R\&D < 0$, or $Emp. \leq 4$ are dropped from estimates.

Table 4—Pre– and Post–2012 Innovation Investment Inputs (Firm-Level)

	asinh(R&D)	asinh(R&D)	asinh(R&D)	R&D direction	asinh(R&D emp.)	asinh(Emp.)
	(1)	(2)	(3)	(4)	(5)	(6)
$I(2009) \times treat$	0.4974	0.3182	0.6186	−17.65	0.2178	0.0842
	(10%)		(5%)			
$I(2010) \times treat$	0.7813	0.6790	0.5454	6.05	0.1268	0.1586
	(5%)	(5%)				(1%)
$I(2011) \times treat$	1.5244	1.4305	0.9969	Omit	0.2102	0.1074
	(1%)	(1%)	(5%)			(5%)
$I(2012) \times treat$	−0.0436	−0.1025	0.4175	−19.81	−0.0223	0.1283
				(5%)		(1%)
$I(2013) \times treat$	−0.2647	−0.2448	0.5850	−18.88	0.0070	0.0460
			(10%)	(5%)		
$I(2014) \times treat$	−0.3777	−0.4511	0.5336	−23.07	−0.0931	0.0511
			(5%)	(1%)		
$I(2015) \times treat$	−0.9392	−1.0564	0.1413	−25.13	−0.2902	0.1547
	(1%)	(1%)		(5%)	(10%)	(1%)
$I(2016) \times treat$	−0.6990	−0.8110	0.2335	−28.70	−0.1703	0.1500
	(10%)	(5%)		(1%)		(1%)
$I(2009) \times size \times treat$	0.0857	0.0986	−0.0048	1.09	0.0017	−0.0067
	(1%)	(1%)				
$I(2010) \times size \times treat$	0.1058	0.1129	0.0178	−0.57	0.0153	−0.0106
	(1%)	(1%)				(5%)
$I(2011) \times size \times treat$	0.0517	0.0568	−0.0013	Omit	0.0167	−0.0093
	(10%)	(10%)				(5%)
$I(2012) \times size \times treat$	0.1377	0.1406	0.0357	1.35	0.0291	−0.0091
	(1%)	(1%)		(5%)	(5%)	(10%)
$I(2013) \times size \times treat$	0.1259	0.1243	0.0109	1.30	0.0245	−0.0085
	(1%)	(1%)		(10%)	(5%)	(10%)
$I(2014) \times size \times treat$	0.1209	0.1268	0.0068	1.60	0.0290	−0.0076
	(1%)	(1%)		(5%)	(10%)	

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Table 4 – *Continued from previous page*

	asinh(R&D)	asinh(R&D)	asinh(R&D)	R&D direction	asinh(R&D emp.)	asinh(Emp.)
	(1)	(2)	(3)	(4)	(5)	(6)
$I(2015) \times size \times treat$	0.1570	0.1666	0.0405	1.79	0.0456	−0.0174
	(1%)	(1%)	(10%)	(5%)	(1%)	(1%)
$I(2016) \times size \times treat$	0.1356	0.1423	0.0333	2.05	0.0314	−0.0156
	(1%)	(1%)		(1%)	(10%)	(1%)
asinh$(Max.\ estab.\ age)$		−0.1685	0.1140	−6.72	0.0695	0.1588
		(0.0926)	(0.0878)	(5.86)	(0.0515)	(0.0602)
asinh$^2(Max.\ estab.\ age)$		0.0226	−0.0154	0.86	−0.0012	−0.0032
		(0.0187)	(0.0234)	(1.15)	(0.0103)	(0.0125)
# clusters	80	80	80	60	80	80
Adj. R^2	0.625	0.629	0.804	0.413	0.778	0.954
Obs.	135,000	130,000	91,000	17,500	102,000	130,000

Note: Estimates adhere to Census requirements. Triple difference coefficients and standard errors are estimated as in (2) using BRDIS panel data from 2008–2016, unless otherwise noted below. The base year is 2008. Year by size coefficients are omitted for tractability. All regressions include firm fixed effects and mode six-digit NAICS by year fixed effects. Standard errors are displayed where available, clustered at the three-digit NAICS level, and calculated from disclosed t-statistics. When standard errors are unavailable, statistical significance is displayed in parentheses and denoted at the missing (i.e., insignificant), 10%, 5%, 1%, levels, to conform with Census. Regressands in Columns (1)–(6) are defined in "Variables." Column (3) is a robustness check on Columns (1) and (2) and limits the sample by an internal Census form completion variable. Column (4) is limited to BRDI-1 form respondents. Where coefficients are missing, years are purposely skipped due to the question not being asked in a specific year. The variable *Size* is the inverse hyperbolic sine of a firm's last known net sales and revenue from 2011 or earlier, and is taken to be zero if no value is reported. The variable *Treat* is an indicator that equals one if a firm's last known response in 2011 or earlier, was that utility patents were "Somewhat" or "Very" important and zero otherwise (i.e., patents are "Not important"). The $I(YYYY)$ variables are dummy variables that are equal to one if the year is equal to $YYYY$ and zero otherwise. Firm-year obs. with $Net\ Sales\ and\ Rev. < 0$, $R\&D < 0$, or $Emp. \leq 4$ are dropped.

Table 5—Pre– and Post–2012 Innovation Outputs (Firm-Level)

	Patent applications	Percent patents applied internationally	Percent sales new products/services	Percent sales new products/services
	(1)	(2)	(3)	(4)
$I(2009) \times treat$	0.3585	−0.16	Base	
$I(2010) \times treat$	0.6476	−10.67	−2.74	
$I(2011) \times treat$	0.1953	14.61	−3.25	
$Post\text{–}2012 \times treat$				−17.65
				(2.68)
$I(2012) \times treat$	0.5818	−9.14	Omit	
$I(2013) \times treat$	0.6429	−1.29	Omit	
$I(2014) \times treat$	0.1470	−6.21	−15.17	
			(1%)	
$I(2015) \times treat$	−0.1328	−12.90	−21.87	
			(1%)	
$I(2016) \times treat$	−0.9395	−17.17	−19.66	
		(10%)	(1%)	
$I(2009) \times size \times treat$	−0.0210	−0.08	Base	
$I(2010) \times size \times treat$	−0.0482	1.01	0.28	
$I(2011) \times size \times treat$	0.0213	−0.81	0.43	
$Post\text{–}2012 \times size \times treat$				1.31
				(0.25)
$I(2012) \times size \times treat$	−0.0451	0.74	Omit	

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Table 5 – *Continued from previous page*

	Patent applications	Percent patents applied internationally	Percent sales new products/services	Percent sales new products/services
	(1)	(2)	(3)	(4)
$I(2013) \times size \times treat$	−0.0624	0.00	Omit	
$I(2014) \times size \times treat$	−0.0339	0.28	1.14	
			(1%)	
$I(2015) \times size \times treat$	−0.0155	0.90	1.70	
			(1%)	
$I(2016) \times size \times treat$	0.0312	1.13	1.59	
			(1%)	
asinh$(Max.\ estab.\ age)$	−0.2828	0.65	−7.67	−7.66
	(0.3404)	(5.45)	(1.81)	(1.80)
asinh$^2(Max\ estab.\ age)$	0.0320	0.19	1.03	1.02
	(0.0616)	(0.95)	(0.36)	(0.36)
# clusters	50	60	80	80
Adj. R^2		0.613	0.480	0.480
Pseudo R^2	0.968			
Obs.	14,500	18,000	39,000	39,000

Note: Estimates adhere to Census requirements. Triple difference coefficients and standard errors are estimated as in (3) using BRDIS panel data from 2008–2016, unless otherwise noted below. The base year is 2008, unless denoted otherwise. Year by size coefficients are omitted for tractability. All regressions include firm fixed effects and mode six-digit NAICS by year fixed effects. Standard errors are displayed where available, clustered at the three-digit NAICS level, and calculated from disclosed t-statistics. When standard errors are unavailable, statistical significance is displayed in parentheses and denoted at the missing (i.e., insignificant), 10%, 5%, 1%, levels, to conform with Census. Regressands are defined in "Variables." Columns (2)–(4) are linear regressions. Column (1) is a Poisson regression. Columns (1)–(2) are limited to BRDI-1 form respondents. Column (3) and (4)'s question was not asked in the 2008 survey and was not asked on all survey forms in 2012 and 2013. Thus, observations from 2008, 2012 and 2013 are not included in Columns (3) and (4). The variable *Size* is the inverse hyperbolic sine of a firm's last known net sales and revenue from 2011 or earlier, and is taken to be zero if no value is reported. The variable *Treat* is an indicator that equals one if a firm's last known response in 2011 or earlier, is that utility patents are "Somewhat" or "Very" important and zero otherwise (i.e., patents are "not important"). The variables $I(YYYY)$ are dummy variables that are equal to one if the year is equal to $YYYY$ and zero otherwise. Firm-year obs. with $Net\ Sales\ and\ Rev. < 0$, $R\&D < 0$, or $Emp. \leq 4$ are dropped.